

Haikong Lu

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Education

The University of Hong Kong

2024.09 – 2028.06

Bachelor of Arts and Sciences in Applied Artificial Intelligence

Double Major in Computer Science

- CGPA: 4.07/4.30; Dean's Honour List (Year 1 & Year 2)
- Research Interests: Reliable AI Systems, Computer-Use Agents, Vision-Language Models, AI Agent Evaluation, Long-Horizon Reasoning

Research Experience

HKU XLang Lab

2026.03 – Present

Undergraduate Research Assistant

Supervisor: Prof. Tao Yu

- Contributing to **OSWorld V2**, a benchmark for evaluating computer-use agents on long-horizon real-world tasks.
- Designing and implementing realistic evaluation tasks to assess agent planning, interaction, and reliability.
- Investigating failure modes of multimodal agents in complex computer environments, focusing on grounding and long-horizon execution.

Publications

OSWorld V2: Benchmarking Computer-Use Agents on Long-Horizon Real-World Tasks

Mengqi Yuan*, Zilong Zhou*, Xinzhuang Xiong*, Weiming Wu, Jiayang Sun, Jiamin Song, Kaiqian Cui, Bowen Wang, Haoyuan Wu, Yitong Li, Dunjie Lu, Haikong Lu, et al.

arXiv, 2026

Selected Projects

Decoder-only Transformer Implementation

PyTorch, Python

- Implemented a decoder-only Transformer from scratch, including RoPE, RMSNorm, SwiGLU, and KV Cache.
- Developed a BPE tokenizer and inference pipeline with temperature-based sampling to study LLM architecture, tokenization, and decoding behaviors.

Academic Research RAG Pipeline

FAISS, Sentence-Transformers

- Built a two-stage retrieval-augmented generation system combining dense retrieval with Cross-Encoder reranking.
- Designed semantic chunking and vector indexing strategies for long-form academic document analysis.

Hearth – LLM-powered Application for Intergenerational Communication

Bronze Prize, Hong Kong Inter-University Generative Artificial Intelligence Hackathon

- Developed an LLM-powered application integrating dialogue generation and backend orchestration.

Skills

AI & Machine Learning: PyTorch, Transformers, LLM Agents, RAG, FAISS, Sentence-Transformers

LLM Systems: Transformer Architecture, RoPE, RMSNorm, BPE Tokenization, KV Cache, Decoding Strategies

Programming: Python, C++, SQL, Shell Scripting

Tools: Linux, Git, Docker, LaTeX

Languages: Mandarin (Native), English (Fluent; TOEFL iBT 111/120)

Additional Experience

Deutsche Börse Group

2026.01 – 2026.02

Finance & Project Intern

Beijing Representative Office

- Produced analytical reports and supported data-driven project analysis for internal strategic discussions.